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Needle Actuator Assembly for Drug Delivery System

Abstract

A needle actuator assembly for a drug delivery system includes a needle actuator body having a guide surface, a needle shuttle having a cam surface, with the needle shuttle moveable along a vertical axis between a first position and a second position. The needle shuttle is configured to move between the first and second position through engagement between the guide surface of the needle actuator body and the cam surface of the needle shuttle. The assembly further including a needle received by the needle shuttle.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application is a continuation of U.S. patent application Ser. No. 18/134,841, Apr. 14, 2023, which is a continuation of U.S. patent application Ser. No. 16/805,996, filed Mar. 2, 2020 (now U.S. Pat. No. 11,654,248), which is a continuation of U.S. patent application Ser. No. 15/616,212, filed Jun. 7, 2017 (now U.S. Pat. No. 10,603,445), which claims priority to U.S. Provisional Application Ser. No. 62/347,921, filed Jun. 9, 2016, the entire disclosures of which are hereby incorporated by reference in their entirety.

BACKGROUND OF THE INVENTION

Field of the Disclosure

[0002] The present disclosure relates generally to an injector device and method for delivering a fluid into the body of a patient by injection.

Description of the Related Art

[0003] Various types of automatic injection devices have been developed to allow drug solutions and other liquid therapeutic preparations to be administered by untrained personnel or to be self-injected. Generally, these devices include a reservoir that is pre-filled with the liquid therapeutic preparation, and some type of automatic needle-injection mechanism that can be triggered by the user. When the volume of fluid or drug to be administered is generally below a certain volume, such as 1 mL, an auto-injector is typically used, which typically has an injection time of about 10 to 15 seconds. When the volume of fluid or drug to be administered is above 1 mL, the injection time generally becomes longer resulting in difficulties for the patient to maintain contact between the device and the target area of the patient's skin. Further, as the volume of drug to be administered becomes larger, increasing the time period for injection becomes desirable. The traditional method for a drug to be injected slowly into a patient is to initiate an IV and inject the drug into the patient's body slowly. Such a procedure is typically performed in a hospital or outpatient setting.

[0004] Certain devices allow for self-injection in a home setting and are capable of gradually injecting a liquid therapeutic preparation into the skin of a patient. In some cases, these devices are small enough (both in height and in overall size) to allow them to be “worn” by a patient while the liquid therapeutic preparation is being infused into the patient. These devices typically include a pump or other type of discharge mechanism to force the liquid therapeutic preparation to flow out of a reservoir and into the injection needle. Such devices also typically include a valve or flow control mechanism to cause the liquid therapeutic preparation to begin to flow at the proper time and a triggering mechanism to initiate the injection.

SUMMARY OF THE INVENTION

[0005] In one aspect, a needle actuator assembly for a drug delivery system includes a needle actuator body having a guide surface, a needle shuttle having a cam surface, with the needle shuttle moveable along a vertical axis between a first position and a second position. The needle shuttle is configured to move between the first and second position through engagement between the guide

surface of the needle actuator body and the cam surface of the needle shuttle. The assembly further including a needle received by the needle shuttle.

[0006] The needle shuttle may include a biasing member configured to move the needle shuttle from the first position to the second position with the guide surface of the needle actuator body disengaged from the cam surface of the needle shuttle. The assembly may include a guide post, where the needle shuttle moves along the guide post. The guide post may be linear. The cam surface of the needle shuttle may include a first cam member and a second cam member spaced from the first cam member, where the guide surface is non-linear and includes a first side and a second side positioned opposite from the first side. The second cam member of the needle shuttle may be configured to engage the second side of the guide surface to move the needle shuttle from the first position to the second position, where the first cam member of the needle shuttle is configured to engage the first side of the guide surface to move the needle shuttle from the second position to the first position.

[0007] In a further aspect, a drug delivery system for injecting a medicament includes a housing, a needle actuator assembly received within the housing, where the needle actuator assembly includes a needle actuator body having a guide surface, with the needle actuator body moveable between a first position and a second position, and a needle shuttle having a cam surface. The needle shuttle is moveable along a vertical axis between a first position and a second position, where the needle shuttle is configured to move between the first and second position through engagement between the guide surface of the needle actuator body and the cam surface of the needle shuttle. The needle actuator assembly also includes a needle received by the needle shuttle. The system further including a button actuator at least partially received by the housing, where movement of the button actuator is configured to move the needle actuator body from the first position to the second position, and where movement of the needle actuator body from the first position to the second position is configured to cause movement of the needle shuttle from the first position to the second position.

[0008] The needle shuttle may include a biasing member configured to move the needle shuttle from the first position to the second position with the guide surface of the needle actuator body disengaged from the cam surface of the needle shuttle. The system may include a guide post extending from the housing, where the needle shuttle moves along the guide post. The guide post may extend about perpendicular from the housing.

[0009] The system may include a pad configured to engage the needle when the needle actuator is in the second position. The pad may be received by a pad arm having a cam surface configured to engage a corresponding cam track on the housing to move the pad beneath the needle.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] The above-mentioned and other features and advantages of this disclosure, and the manner of attaining them, will become more apparent and the disclosure itself will be better understood by reference to the following descriptions of embodiments of the disclosure taken in conjunction with the accompanying drawings, wherein:

[0011] FIG. 1 is a perspective view of a drug delivery system according to one aspect of the present invention.

[0012] FIG. 2 is a perspective, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention.

[0013] FIG. 3 is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention.

[0014] FIG. 4 is a top view of the drug delivery system of FIG. 1 according to one aspect of the

present invention, showing a top portion of the housing removed and the drug delivery system in a pre-use position.

[0015] FIG. 5 is a top, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a pre-use position.

[0016] FIG. 6 is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a pre-use position.

[0017] FIG. 7 is a top view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a top portion of the housing removed and the drug delivery system in an initial actuation position.

[0018] FIG. 8 is a top, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in an initial actuation position.

[0019] FIG. 9 is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in an initial actuation position.

[0020] FIG. 10 is a top view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a top portion of the housing removed and the drug delivery system in a use position.

[0021] FIG. 11 is a top, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a use position.

[0022] FIG. 12 is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a use position.

[0023] FIG. 13 is a top view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a top portion of the housing removed and the drug delivery system in a post-use position.

[0024] FIG. 14 is a top, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a post-use position.

[0025] FIG. 15 is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing the drug delivery system in a post-use position.

[0026] FIG. 15A is a front, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a pad with the drug delivery system in a pre-use position.

[0027] FIG. 15B is a perspective, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a pad with the drug delivery system in a pre-use position.

[0028] FIG. 15C is a perspective, cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a pad with the drug delivery system in a pre-use position.

[0029] FIG. 16 is a partial cross-sectional view of the drug delivery system of FIG. 1 according to one aspect of the present invention, showing a valve assembly.

[0030] FIG. 17 is a perspective view of a drive assembly for a drug delivery system according to one aspect of the present invention.

[0031] FIG. 18 is a cross-sectional view of the drive assembly of FIG. 17 according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0032] FIG. 19 is a cross-sectional view of the drive assembly of FIG. 17 according to one aspect of the present invention, showing a use position of the drive assembly.

[0033] FIG. 20 is a cross-sectional view of the drive assembly of FIG. 17 according to one aspect of the present invention, showing a post-use position of the drive assembly.

[0034] FIG. 21 is a perspective view of a plunger actuation member of the drive assembly of FIG. 17 according to one aspect of the present invention.

[0035] FIG. 22 is a perspective view of a first plunger member of the drive assembly of FIG. 17

according to one aspect of the present invention.

[0036] FIG. **23** is a perspective view of a plunger actuation member and first plunger member of the drive assembly of FIG. **17** according to one aspect of the present invention, showing the plunger actuation member engaged with the first plunger member.

[0037] FIG. **24** is a perspective view of a plunger actuation member and first plunger member of the drive assembly of FIG. **17** according to one aspect of the present invention, showing the plunger actuation member disengaged from the first plunger member.

[0038] FIG. **25** is a perspective view of a plunger actuation member and first plunger member of the drive assembly of FIG. **17** according to one aspect of the present invention, showing the plunger actuation member disengaged from and axially displaced relative to the first plunger member.

[0039] FIG. **26** is a front view of a first plunger member and a second plunger member of the drive assembly of FIG. **17** according to one aspect of the present invention.

[0040] FIG. **27** is a top view of a drive assembly for a drug delivery system according to a further aspect of the present invention.

[0041] FIG. **28** is a perspective view of the drive assembly of FIG. **27** according to one aspect of the present invention.

[0042] FIG. **29** is a cross-sectional view of the drive assembly of FIG. **27** according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0043] FIG. **30** is a perspective view of the drive assembly of FIG. **27** according to one aspect of the present invention, showing the drive assembly received by a bottom portion of a housing.

[0044] FIG. **31** is a perspective view of the housing of FIG. **30** according to one aspect of the present invention.

[0045] FIG. **32** is a top view of the drive assembly of FIG. **27** according to one aspect of the present invention, showing engagement of the drive assembly with a portion of a needle actuator in an initial actuation position of the drive assembly.

[0046] FIG. **33** is an enlarged perspective view of the drive assembly of FIG. **27** according to one aspect of the present invention, showing engagement of the drive assembly with a portion of a needle actuator in an initial actuation position of the drive assembly.

[0047] FIG. **34** is a front view of a needle actuator assembly according to one aspect of the present invention.

[0048] FIG. **35** is a left side perspective view of a needle shuttle of the needle actuator assembly of FIG. **34** according to one aspect of the present invention.

[0049] FIG. **36** is a right side perspective view of a needle shuttle of the needle actuator assembly of FIG. **34** according to one aspect of the present invention.

[0050] FIG. **37A** is a front view of the needle actuator assembly of FIG. **34** according to one aspect of the present invention, showing the needle actuator assembly in a pre-use position.

[0051] FIG. **37B** is a front view of the needle actuator assembly of FIG. **34** according to one aspect of the present invention, showing the needle actuator assembly in a use position.

[0052] FIG. **37C** is a front view of the needle actuator assembly of FIG. **34** according to one aspect of the present invention, showing the needle actuator assembly in an initial post-use position.

[0053] FIG. **37D** is a front view of the needle actuator assembly of FIG. **34** according to one aspect of the present invention, showing the needle actuator assembly in a post-use position.

[0054] FIG. **38A** is a perspective view of the needle actuator assembly of FIG. **34** according to one aspect of the present invention, showing the needle actuator assembly in a use position.

[0055] FIG. **38B** is a perspective view of the needle actuator assembly of FIG. **34** according to one aspect of the present invention, showing the needle actuator assembly in an initial post-use position.

[0056] FIG. **39** is a perspective view of an actuator button and the needle actuator assembly of FIG. **34** according to one aspect of the present invention, showing the needle actuator assembly in an

initial post-use position.

[0057] FIG. **40A** is a cross-sectional view of an actuator button and the needle actuator assembly of FIG. **34** according to one aspect of the present invention, showing the needle actuator assembly in an initial post-use position.

[0058] FIG. **40B** is a perspective view of an actuator button and the needle actuator assembly of FIG. **34** according to one aspect of the present invention, showing the needle actuator assembly in a post-use position.

[0059] FIG. **41** is a perspective view of a drive assembly for a drug delivery system according to a further aspect of the present invention.

[0060] FIG. **42** is a perspective view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing a top portion of a housing removed.

[0061] FIG. **43** is a cross-sectional view of the drive assembly of FIG. **41** according to one aspect of the present invention.

[0062] FIG. **44** is a perspective view of the drive assembly of FIG. **41** according to one aspect of the present invention.

[0063] FIG. **45** is a cross-sectional view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a pre-use position.

[0064] FIG. **46** is a cross-sectional view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a pre-use position.

[0065] FIG. **47** is a top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a pre-use position.

[0066] FIG. **48** is a top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in an initial actuation position.

[0067] FIG. **49** is a top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in an initial actuation position.

[0068] FIG. **50** is a top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly an initial actuation position.

[0069] FIG. **51** is a top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a use position.

[0070] FIG. **52** is a top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a use position.

[0071] FIG. **53** is a cross-sectional view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a use position.

[0072] FIG. **54** is a top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a use position.

[0073] FIG. **55** is a cross-sectional view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a use position.

[0074] FIG. **56** is a cross-sectional view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a use position.

[0075] FIG. **57** is a top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a use position.

[0076] FIG. **58** is a top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in an initial post-use position.

[0077] FIG. **59** is a perspective view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in an initial post-use position.

[0078] FIG. **60** is a top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a post-use position.

[0079] FIG. **61** top view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a post-use position.

[0080] FIG. **62** is a cross-sectional view of the drive assembly of FIG. **41** according to one aspect

of the present invention, showing the drive assembly in a pre-use position.

[0081] FIG. **63** is a cross-sectional view of the drive assembly of FIG. **41** according to one aspect of the present invention, showing the drive assembly in a use position.

[0082] FIG. **64** is a perspective view of a drive assembly according to a further aspect of the present invention.

[0083] FIG. **65A** is a front view of a needle actuator assembly according to one aspect of the present invention, showing the needle actuator assembly in a use position.

[0084] FIG. **65B** is a front view of the needle actuator assembly of FIG. **65A** according to one aspect of the present invention, showing the needle actuator assembly in a use position.

[0085] FIG. **65C** is a front view of the needle actuator assembly of FIG. **65A** according to one aspect of the present invention, showing the needle actuator assembly in an initial post-use position.

[0086] FIG. **65D** is a front view of the needle actuator assembly of FIG. **65A** according to one aspect of the present invention, showing the needle actuator assembly in a post-use position.

[0087] FIG. **65E** is a front view of the needle actuator assembly of FIG. **65A** according to one aspect of the present invention, showing the needle actuator assembly in a pre-use position.

[0088] FIG. **65F** is a cross-sectional view of the needle actuator assembly of FIG. **65A** according to one aspect of the present invention, showing the needle actuator assembly in a pre-use position.

[0089] FIG. **65G** is a front view of the needle actuator assembly of FIG. **65A** according to one aspect of the present invention, showing the needle actuator assembly in a pre-use position with a button actuator axially displaced.

[0090] FIG. **65H** is a cross-sectional view of the needle actuator assembly of FIG. **65A** according to one aspect of the present invention, showing the needle actuator assembly in a pre-use position with a button actuator axially displaced.

[0091] FIG. **66** is a perspective view of a button spring of the needle actuator assembly of FIG. **65A** according to one aspect of the present invention.

[0092] FIG. **67** is a perspective view of an actuator button of the needle actuator assembly of FIG. **65A** according to one aspect of the present invention.

[0093] FIG. **68** is a cross-sectional view of a button spring and actuator button of the needle actuator assembly of FIGS. **65A** according to one aspect of the present invention.

[0094] FIG. **68A** is a perspective view of an actuator button of the needle actuator assembly of FIG. **65A** according to a further aspect of the present invention.

[0095] FIG. **68B** is a bottom view of an actuator button of the needle actuator assembly of FIG. **65A** according to a further aspect of the present invention.

[0096] FIG. **68C** is a front view of an actuator button of the needle actuator assembly of FIG. **65A** according to a further aspect of the present invention.

[0097] FIG. **68D** is a top view of an actuator button of the needle actuator assembly of FIG. **65A** according to a further aspect of the present invention, showing the actuator button in a pre-use position.

[0098] FIG. **68E** is a front view of an actuator button of the needle actuator assembly of FIG. **65A** according to a further aspect of the present invention, showing the actuator button in a pre-use position.

[0099] FIG. **68F** is a top view of an actuator button of the needle actuator assembly of FIG. **65A** according to a further aspect of the present invention, showing the actuator button in a use position.

[0100] FIG. **68G** is a front view of an actuator button of the needle actuator assembly of FIG. **65A** according to a further aspect of the present invention, showing the actuator button in a use position.

[0101] FIG. **69** is a top view of an actuator button of the needle actuator assembly of FIG. **65A** according to one aspect of the present invention.

[0102] FIG. **70A** is a schematic view of a drive assembly according to one aspect of the present invention, showing the drive assembly in a pre-use position.

[0103] FIG. **70B** is a schematic view of the drive assembly of FIG. **70A** according to one aspect of the present invention, showing the drive assembly in a use position.

[0104] FIG. **70C** is a schematic view of the drive assembly of FIG. **70A** according to one aspect of the present invention, showing the drive assembly in a use position.

[0105] FIG. **70D** is a schematic view of the drive assembly of FIG. **70A** according to one aspect of the present invention, showing the drive assembly in a use position.

[0106] FIG. **70E** is a schematic view of the drive assembly of FIG. **70A** according to one aspect of the present invention, showing the drive assembly in a use position.

[0107] FIG. **70F** is a schematic view of the drive assembly of FIG. **70A** according to one aspect of the present invention, showing the drive assembly in a post-use position.

[0108] FIG. **70G** is a schematic view of the drive assembly of FIG. **70A** according to one aspect of the present invention, showing the drive assembly in a post-use position.

[0109] FIG. **71** is a perspective view of a spacer assembly for a drug delivery system according to one aspect of the present invention, showing an assembled, pre-use position of the spacer assembly.

[0110] FIG. **72** is a perspective view of the spacer assembly of FIG. **71** according to one aspect of the present invention, showing a use position of the spacer assembly.

[0111] FIG. **73** is a perspective view of the spacer assembly of FIG. **71** according to one aspect of the present invention, showing an initial post-use position of the spacer assembly.

[0112] FIG. **74** is a perspective view of a restriction member according to one aspect of the present invention.

[0113] FIG. **75** is a front view of a spacer assembly for a drug delivery system according to a further aspect of the present invention.

[0114] FIG. **76** is a top view of a spacer assembly for a drug delivery system according to one aspect of the present invention.

[0115] FIG. **77** is a perspective view of the spacer assembly of FIG. **76** according to one aspect of the present invention.

[0116] FIG. **78** is a cross-sectional view of the spacer assembly of FIG. **76** according to one aspect of the present invention.

[0117] FIG. **79** is a perspective view of a spacer assembly for a drug delivery system according to a further aspect of the present invention.

[0118] FIG. **80** is a perspective view of a spacer assembly for a drug delivery system according to another aspect of the present invention.

[0119] FIG. **81A** is a cross-sectional view of the spacer assembly of FIG. **80** according to one aspect of the present invention, showing a pre-assembly position of the spacer assembly.

[0120] FIG. **81B** is a cross-sectional view of the spacer assembly of FIG. **80** according to one aspect of the present invention, showing an assembled position of the spacer assembly.

[0121] FIG. **82** is a perspective view of a drive assembly for a drug delivery system according to one aspect of the present invention.

[0122] FIG. **83** is a perspective view of the drive assembly of FIG. **82** according to one aspect of the present invention, showing a top portion of a housing removed.

[0123] FIG. **84** is a cross-sectional view of the drive assembly of FIG. **82** according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0124] FIG. **85** is an enlarged cross-sectional view of the drive assembly of FIG. **82** according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0125] FIG. **86** is a top view of a biasing member of the drive assembly of FIG. **82** according to one aspect of the present invention.

[0126] FIG. **87** is a perspective view of the drive assembly of FIG. **82** according to one aspect of the present invention, showing a restriction member engaged with the drive assembly.

[0127] FIG. **88** is a perspective view of a drive assembly for a drug delivery system according to one aspect of the present invention.

[0128] FIG. **89** is a perspective view of the drive assembly of FIG. **88** according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0129] FIG. **90** is a cross-sectional view of the drive assembly of FIG. **88** according to one aspect of the present invention.

[0130] FIG. **91** is a perspective view of the drive assembly of FIG. **88** according to one aspect of the present invention, showing a post-use position of the drive assembly.

[0131] FIG. **92** is a cross-sectional view of the drive assembly of FIG. **88** according to one aspect of the present invention, showing a pre-use position of the drive assembly.

[0132] FIG. **93** is a front view of the drive assembly of FIG. **88** according to one aspect of the present invention, showing a use position of the drive assembly.

[0133] FIG. **94** is a perspective view of a spacer assembly for a drug delivery system according to one aspect of the present invention.

[0134] FIG. **95** is a front view of the spacer assembly of FIG. **94** according to one aspect of the present invention.

[0135] FIG. **96** is a cross-sectional view of the spacer assembly of FIG. **94** according to one aspect of the present invention.

[0136] FIG. **97** is a perspective view of the spacer assembly of FIG. **94** according to one aspect of the present invention, showing a shim removed.

[0137] FIG. **98** is a perspective view of a fixed spacer of the spacer assembly of FIG. **94** according to one aspect of the present invention.

[0138] FIG. **99** is a perspective view of an adjustable spacer of the spacer assembly of FIG. **94** according to one aspect of the present invention.

[0139] FIG. **100** is a perspective view of a shim of the spacer assembly of FIG. **94** according to one aspect of the present invention.

[0140] Corresponding reference characters indicate corresponding parts throughout the several views. The exemplifications set out herein illustrate exemplary aspects of the disclosure, and such exemplifications are not to be construed as limiting the scope of the disclosure in any manner.

DETAILED DESCRIPTION

[0141] The following description is provided to enable those skilled in the art to make and use the described embodiments contemplated for carrying out the invention. Various modifications, equivalents, variations, and alternatives, however, will remain readily apparent to those skilled in the art. Any and all such modifications, variations, equivalents, and alternatives are intended to fall within the spirit and scope of the present invention.

[0142] For purposes of the description hereinafter, the terms “upper”, “lower”, “right”, “left”, “vertical”, “horizontal”, “top”, “bottom”, “lateral”, “longitudinal”, and derivatives thereof shall relate to the invention as it is oriented in the drawing figures. However, it is to be understood that the invention may assume various alternative variations, except where expressly specified to the contrary. It is also to be understood that the specific devices illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the invention. Hence, specific dimensions and other physical characteristics related to the embodiments disclosed herein are not to be considered as limiting.

[0143] Referring to FIGS. **1-16**, a drug delivery system **10** according to one aspect of the present invention includes a drive assembly **12**, a container **14**, a valve assembly **16**, and a needle actuator assembly **18**. The drive assembly **12**, the container **14**, the valve assembly **16**, and the needle actuator assembly **18** are at least partially positioned within a housing **20**. The housing **20** includes a top portion **22** and a bottom portion **24**, although other suitable arrangements for the housing **20** may be utilized. In one aspect, the drug delivery system **10** is an injector device configured to be worn or secured to a user and to deliver a predetermined dose of a medicament provided within the container **14** via injection into the user. The system **10** may be utilized to deliver a “bolus injection” where a medicament is delivered within a set time period. The medicament may be delivered over a

time period of up to 45 minutes, although other suitable injection amounts and durations may be utilized. A bolus administration or delivery can be carried out with rate controlling or have no specific rate controlling. The system **10** may deliver the medicament at a fixed pressure to the user with the rate being variable. The general operation of the system **10** is described below in reference to FIGS. **1-16** with the specifics of the drive assembly **12**, needle actuator assembly **18**, and other features of the system **10**, discussed below in connection with FIGS. **17-93**.

[0144] Referring again to FIGS. **1-16**, the system **10** is configured to operate through the engagement of an actuation button **26** by a user, which results in a needle **28** of the needle assembly **18** piercing the skin of a user, the actuation of the drive assembly **12** to place the needle **28** in fluid communication with the container **14** and to expel fluid or medicament from the container **14**, and the withdrawal of the needle **28** after injection of the medicament is complete. The general operation of a drug delivery system is shown and described in International Publication Nos. 2013/155153 and 2014/179774, which are hereby incorporated by reference in their entirety. The housing **20** of the system **10** includes an indicator window **30** for viewing an indicator arrangement **32** configured to provide an indication to a user on the status of the system **10** and a container window **31** for viewing the container **14**. The indicator window **30** may be a magnifying lens for providing a clear view of the indicator arrangement **32**. The indicator arrangement **32** moves along with the needle actuator assembly **18** during use of the system **10** to indicate a pre-use status, use status, and post-use status of the system **10**. The indicator arrangement **32** provides visual indicia regarding the status, although other suitable indicia, such as auditory or tactile, may be provided as an alternative or additional indicia.

[0145] Referring to FIGS. **4-6**, during a pre-use position of the system **10**, the container **14** is spaced from the drive assembly **12** and the valve assembly **16** and the needle **28** is in a retracted position. During the initial actuation of the system **10**, as shown in FIGS. **7-9**, the drive assembly **12** engages the container **14** to move the container **14** toward the valve assembly **16**, which is configured to pierce a closure **36** of the container **14** and place the medicament within the container **14** in fluid communication with the needle **28** via a tube (not shown) or other suitable arrangement. The drive assembly **12** is configured to engage a stopper **34** of the container **14**, which will initially move the entire container **14** into engagement with the valve assembly **16** due to the incompressibility of the fluid or medicament within the container **14**. The initial actuation of the system **10** is caused by engagement of the actuation button **26** by a user, which releases the needle actuator assembly **18** and the drive assembly **12** as discussed below in more detail. During the initial actuation, the needle **28** is still in the retracted position and about to move to the extended position to inject the user of the system **10**.

[0146] During the use position of the system **10**, as shown in FIGS. **10-12**, the needle **28** is in the extended position at least partially outside of the housing **20** with the drive assembly **12** moving the stopper **34** within the container **14** to deliver the medicament from the container **14**, through the needle **28**, and to the user. In the use position, the valve assembly **16** has already pierced a closure **36** of the container **14** to place the container **14** in fluid communication with the needle **28**, which also allows the drive assembly **12** to move the stopper **34** relative to the container **14** since fluid is able to be dispensed from the container **14**. At the post-use position of the system **10**, shown in FIGS. **13-15**, the needle **28** is in the retracted position and engaged with a pad **38** to seal the needle **28** and prevent any residual flow of fluid or medicament from the container **14**. The container **14** and valve assembly **16** may be the container **14** and valve assembly **16** shown and described in International Publication No. WO 2015/081337, which is hereby incorporated by reference in its entirety.

[0147] Referring to FIGS. **15A-15C**, the pad **38** is biased into the pad as the needle actuator body **96** moves from the use position to the post-use position. In particular, the pad **38** is received by a pad arm **122** having a cam surface **124** that cooperates with a cam track **126** on the bottom portion **24** of the housing **20**. The pad arm **122** is connected to the needle actuator body **96** via a torsion bar

128. The cam surface **124** is configured to engage the cam track **126** to deflect the pad arm **122** downwards thereby allowing the pad **38** to pass beneath the needle **28** before being biased upwards into the needle **28**. The torsion bar **128** allows the pad arm **122** to twist about a pivot of the needle actuator body **96**. The pad **38** may be press-fit into an opening of the pad arm **122**, although other suitable arrangements for securing the pad **38** may be utilized.

[0148] Referring to FIGS. **1-33**, the drive assembly **12** according to one aspect of the present invention is shown. As discussed above, the drive assembly **12** is configured to move the container **14** to pierce the closure **36** of the container **14** and also to move the stopper **34** within the container **14** to dispense fluid or medicament from the container **14**. The drive assembly **12** shown in FIGS. **17-33** is configured to engage and cooperate with a spacer assembly **40** received by the stopper **34** of the container **14**. The spacer assembly **40** includes a spacer **42** and a spacer holder **44**. The spacer holder **44** is received by the stopper **34** and the spacer **42** is received by the spacer holder **44**. The spacer holder **44** includes a first threaded portion **46** that engages a corresponding threaded portion of the stopper **34**, although other suitable arrangements may be utilized. The spacer **42** also includes a threaded portion **48** that engages a corresponding second threaded portion **50** of the spacer holder **44** for securing the spacer **42** to the spacer holder **44**, although other suitable arrangements may be utilized. The drive assembly **12** is configured to dispense a range of pre-determined fill volumes of the container **14** while maintaining the functional features of the system **10** described above, including, but not limited to, retraction of the needle **28** after the end of the dose and providing an indication of the status of the system **10** while also minimizing abrupt engagement of the stopper **34** by the drive assembly **12**. The drive assembly **12** is configured to dispense a plurality of discrete fill volume ranges by utilizing a plurality of sizes of the spacers **42**. In one aspect, twelve fill volume ranges and twelve spacer **42** sizes are provided. In one aspect, the length of the spacer **42** is changed to accommodate different fill volumes in the container **14**. Alternatively, a single size spacer **42** may be utilized with a plurality of fill volumes in the container **14** accommodated by utilizing a plurality of shims that are received by the spacer **42**.

[0149] Referring to FIGS. **17-26**, the drive assembly **12** includes a first plunger member **52**, a second plunger member **54** received by the first plunger member **52**, a first biasing member **56**, a second biasing member **58**, a plunger actuation member **60**, and an index member **62**. The first plunger member **52** is moveable from a pre-use position (shown in FIG. **18**), to a use position (shown in FIG. **19**), to a post-use position (shown in FIG. **20**) with the first plunger member **52** configured to engage the spacer assembly **40** and move the stopper **34** within the container **14** to dispense medicament from the container **14**. The first plunger member **52** is configured to move axially. The second plunger member **54** and the first plunger member **52** form a telescoping arrangement with the second plunger **54** configured to move axially after the first plunger member **52** moves a predetermined axial distance. The movement of the first and second plunger members **52**, **54** is provided by the first and second biasing members **56**, **58**, which are compression springs, although other suitable arrangements for the biasing members **56**, **58** may be utilized.

[0150] The first biasing member **56** is received by the second plunger member **54** and is constrained between the plunger actuation member **60** (and index member **62**) and a first spring seat **64** of the second plunger member **54**. The second biasing member **58** is positioned radially inward from the first biasing member **56** and received by the second plunger member **54**. The second biasing member **58** is constrained between a second spring seat **66** of the second plunger member **54** and the first plunger member **52**. The second biasing member **58** is configured to bias the first plunger **52** member towards the container **14** from the pre-use position, to the use position, and to the post-use position. The first biasing member **56** is configured to bias the second plunger member **54** towards the container **14**, which, in turn, biases the first plunger member **52** towards the container **14** from the pre-use position, to the use position, and to the post-use position. More specifically, the second biasing member **58** is configured to drive the first plunger member **52** against the spacer assembly **40** or stopper **34** to move the container **14** into engagement the valve

assembly **16** thereby piercing the closure **36** of the container **14** and placing the container **14** in fluid communication with the needle **28**. The first biasing member **56** is configured to move the stopper **34** within the container **14** to dispense the medicament within the container **14**. The second biasing member **58** has a different spring constant than the first biasing member **56**. In particular, the second biasing **58** member is stiffer than the first biasing member **56** to provide a high force for piercing the closure **36** of the container **14** while the first biasing member **56** provides a lower force for dispensing as appropriate for the viscosity of the fluid or medicament within the container **14**. [0151] Referring again to FIGS. **17-26**, the plunger actuation member **60** has an annular portion **68** and a spindle portion **70**. The plunger actuation member **60** is rotationally moveable relative to the first plunger member **52** between a first rotational position and a second rotational position spaced from the first rotational position. The first rotational position may be 15 degrees from the second rotational position, although other suitable positions may be utilized. The annular portion **68** includes a drive surface **72** including a plurality of gears **74**, although other suitable arrangements may be utilized for the drive surface **72**. The spindle portion **70** includes an actuator locking surface **76** configured for engagement and release from a plunger locking surface **78** of the first plunger member **52**. The plunger locking surface **78** includes a plurality of projections **80** configured to be received by a plurality of slots or cutouts **81** defined by the actuator locking surface **76**.

[0152] As shown in FIGS. **18** and **23**, in the first rotational position of the plunger actuation member **60**, the plurality of projections **80** and the plurality of slots or cutouts **81** are out of alignment such that the plunger actuation member **80** is engaged with the first plunger member **52** to prevent movement of the first and second plunger members **52**, **54** with the first and second biasing members **56**, **58** biasing the first and second plunger members **52**, **54** away from the plunger actuation member **60**. As shown in FIGS. **19** and **24**, in the second rotational position of the plunger actuation member **60**, the plurality of projections **80** and the plurality of slots or cutouts **81** are aligned with each other such that the plunger actuation member **60** is disengaged with the first plunger member **52** to allow movement of the first and second plunger members **52**, **54** thereby starting the dispensing process from the container **14**.

[0153] Referring to FIGS. **7** and **33**, the drive surface **72** of the plunger actuation member **60** is configured to be engaged by a portion of the needle actuator assembly **18**. After engagement of the actuator button **26** and release of the needle actuator assembly **18**, which is discussed in more detail below, the needle actuator assembly **18** moves within the housing **20** from the pre-use position, to the use position, and to the post-use position. During the initial movement of the needle actuator assembly **18**, a portion of the needle actuator assembly **18** engages the drive surface **72** of the plunger actuation member **60** to move the plunger actuation member **60** from the first rotational position to the second rotational position. As shown in FIG. **33**, an angled blade portion **82** of the needle actuator assembly **18** engages the drive surface **72** of the plunger actuation member **60** to cause rotation of the plunger actuation member **60**.

[0154] Referring to FIGS. **11**, **13**, and **26**, the second plunger member **52** includes a plurality of coded projections **84** with a preselected one of the plurality of coded projections **84** configured to engage a restriction member **86** of the system **10**. As discussed in more detail below, the restriction member **86** cooperates with the needle actuation assembly **18** and restricts movement of the needle actuator assembly **18** from the use position to the post-use position until a predetermined end-of-dose position of the stopper **34** is reached. In one aspect, the restriction member **86** is configured to restrict axial movement of the needle actuation assembly **18** from the use position through engagement between the restriction member **86** and a portion of the needle actuation assembly **18**. Such engagement between the restriction member **86** and the needle actuation assembly **18** is released by rotation of the restriction member **86** when the stopper **34** reaches the end-of-dose position. During the use position of the needle actuator assembly **18**, the restriction member **86** is biased in a rotational direction with the rotation of the restriction member **86** being prevented

through engagement between the restriction member **86** and one of the plurality of coded projections **84** of the second plunger member **54**. The plurality of coded projections **84** may be axial ribs of varying length, although other suitable arrangements may be utilized. Each coded projection **84** defines a point at which the restriction member **86** is able to rotate thereby releasing the needle actuator assembly **18**. The smooth portion of the second plunger member **52** may also provide a further “code” for determining when the system **10** transitions to the end-of-dose position.

[0155] As discussed above, the indicator arrangement **32** moves with different portions of the indicator arrangement **32** visible through the indicator window **30** as the system **10** moves from the pre-use, use, and post-use or end-of-dose positions. More specifically, the indicator arrangement **32** engages a portion of the restriction member **86** and moves along with the restriction member **86** through the various stages of the system **10** to provide an indication to the user regarding the state of the system **10**.

[0156] During assembly of the system **10**, the dosage of the container **14** is matched with a specific spacer **42** having a set length and a corresponding one of the plurality of coded projections **84** is aligned with the restriction member **86**. Accordingly, as discussed above, the container **14** may be provided with a plurality of dosage volumes with each volume corresponding to a specific spacer **42** and coded projection **84**. Thus, even for different dosage volumes, the system **10** is configured to inject the needle **28** into the user to deliver a dose of medicament from the container **14**, retract the needle **28** after the end of the dose, and provide an indication of the status of the system **10** while minimizing abrupt engagement of the stopper **34** by the drive assembly **12**. In particular, the size of the stopper **34** may be selected to minimize the distance between the first plunger member **52** and the spacer assembly **40** and does not require the use of damping.

[0157] Referring to FIGS. **27-33**, a drive assembly **12A** according to a further aspect of the present invention is shown. The drive assembly **12A** shown in FIGS. **27-33** is similar to and operates in the same manner as the drive assembly **12** shown in FIGS. **17-26** and described above. In the drive assembly of FIGS. **27-33**, however, the first plunger member **52** is received by the second plunger member **54** and extends from the second plunger member **54** during axial movement from the pre-use position to the use position. Further, the first plunger member **52** includes an extension portion **88** configured to engage the second plunger member **54** after the first plunger member **52** moves predetermined axial distance such that the first and second plunger members **52**, **54** move together. The first and second biasing members **56**, **58** engage and act on the first and second plunger members **52**, **54** in the same manner as the drive assembly **12** of FIGS. **17-26**.

[0158] Referring to FIGS. **27-32**, the index member **62** is positioned about the first and second plunger members **52**, **54** and includes a plurality of ratchet teeth **90** configured to engage a flexible tab **92** positioned on the bottom portion **24** of the housing **20**. When the drive assembly **12**, **12A** is installed into the bottom portion **24** of the housing **20**, the engagement of the ratchet teeth **90** of the index member **62** with the flexible tab **92** of the housing **20** provide a one-way rotation of the index member **62**. The index member **62** is configured to rotate to align one of the coded projections **84** of the second plunger member **52** with the restriction member **86** based on the dosage volume and spacer **42** size as discussed above. The index member **62** may provide the drive assembly **12**, **12A** with **24** rotational positions of which **12** may have unique dose values associated with them.

[0159] Referring to FIGS. **1-16** and **34-40B**, the needle actuator assembly **18** according to one aspect of the present invention is shown. The needle actuator assembly **18** includes a needle actuator body **96** having guide surfaces **98**, a needle shuttle **102** having cam surfaces **104**, and the needle **28** received by the needle shuttle **102** and configured to be in fluid communication with the container **14** as discussed above. The needle actuator body **96** is generally rectangular with the guide surfaces **98** protruding radially inward. The needle shuttle **102** is received within the needle actuator body **96**. As described above, the needle actuator body **96** is moveable within the housing **20** from a pre-use position (shown in FIGS. **4-6**), an initial actuation position (FIGS. **7-9**), a use

position (FIGS. 10-12), and a post-use position (FIGS. 13-15). The needle actuator body **96** is biased from the pre-use position to the post-use position via an extension spring **106**, although other suitable biasing arrangements may be utilized. The needle actuator body **96** is released and free to move from the pre-use position to the use position upon engagement of the actuator button **26**, which is discussed in more detail below. The needle actuator body **96** moves from the use position to the post-use position after rotation of the restriction member **86** as discussed above in connection with FIGS. 17-33.

[0160] Referring to FIGS. 34-40B, the needle shuttle **102** is moveable along a vertical axis between a retracted position where the needle **28** is positioned within the housing **20** and an extended position where at least a portion of the needle **28** extends out of the housing **20**. The needle shuttle **102** is configured to move between the retracted position and the extended position through engagement between the guide surfaces **98** of the needle actuator **96** and the cam surfaces **104** of the needle shuttle **102**. The cam surfaces **104** are provided by first and second cam members **108**, **110**, with the first cam member **108** spaced from the second cam member **110**. The housing **20** includes a guide post **112** having recess configured to receive a T-shaped projection **114** on the needle shuttle **102**, although other shapes and configurations may be utilized for the guide post **112** and T-shaped projection **114**. The needle shuttle **102** moves along the guide post **112** between the retracted and extended positions. The guide post **112** is linear and extends about perpendicular from the housing **20**, although other suitable arrangements may be utilized. The guide surfaces **98** of the needle actuator body **86** are non-linear and each include a first side **116** and a second side **118** positioned opposite from the first side **116**.

[0161] As discussed below, the guide surfaces **98** of the needle actuator body **96** cooperate with the cam members **108**, **110** of the needle shuttle **102** to move the needle shuttle **102** vertically between the retracted and extended positions as the needle actuator body **96** moves axially from the pre-use position to the post-use position. The needle shuttle **102** also includes a shuttle biasing member **120** configured to engage the housing **20** or the actuator button **26**. In particular, the shuttle biasing member **120** engages the housing **20** or actuator button **26** and provides a biasing force when the needle actuator body **96** is transitioning from the use position to the post-use position. When the needle actuator body **96** is fully transitioned to the post-use position, the cam members **108**, **110** of the needle shuttle **102** are disengaged from the guide surfaces **98** of the needle actuator body **96** and the shuttle biasing member **120** biases the needle shuttle **102** downward such that the needle **28** engages the pad **38**, as discussed above. As discussed above in connection with FIGS. 1-16, however, the pad **38** may also be biased into the needle **28** rather than biasing the needle shuttle **102** downwards via the shuttle biasing member **120**. The needle actuator body **96** may interact with the actuator button **26** to prevent the actuator button **26** from popping back up until the post-use position is reached, which is discussed below in more detail.

[0162] Referring to FIGS. 37A-40B, in a pre-use position (FIG. 37A), the needle shuttle **102** is in the retracted position with the cam members **108**, **110** spaced from the guide surface **98** of the needle actuator body **96**. As the needle actuator body **96** moves to the use position (FIGS. 37B and 38A), the second cam member **110** of the needle shuttle **102** engages the second side **118** of the guide surfaces **98** to move the needle shuttle **102** from the retracted position to the extended position. During the transition from the use position to the post-use position of the needle actuator body **96** (FIG. 37C), the first cam member **108** of the needle shuttle **102** is engaged with the first side **116** of the guide surfaces **98** to move the needle shuttle **102** from the second position to the first position. After the needle actuator body **96** is fully transitioned to the post-use position (FIGS. 37D and 38B), the shuttle biasing member **120** biases the needle shuttle **102** downward as the cam members **108**, **110** disengage from the guide surfaces **98** of the needle actuator body **96** with the needle **28** engaging the pad **38**. The transition of the needle actuator body **96** and the corresponding position of the needle shuttle **102** is also shown in FIGS. 39-40B. The interaction between the actuator button **26** and the needle actuator body **96** is discussed in detail in connection with FIGS.

65A-67. Referring to FIGS. 41-64, a drug delivery system **200** according to a further embodiment is shown. The system **200** includes a housing **202** having an upper housing **204** and a lower housing **206**. The housing has a proximal end **205** and a distal end **207**. The upper housing **204** has a status view port **208** so that a user can view the operating status of the system **200**. The system **200** also includes a valve assembly **212**, a tube **214** fluidly connecting the valve assembly **214** with a patient needle **215** that is disposed in a proximal end of a needle arm **216**. A spring **218** biases a needle actuator **220** distally.

[0163] As shown in FIGS. 42-46, the system **200** additionally includes a container or medicament container **222** with a stopper **224** movably disposed therein, although the stopper **224** is omitted from various figures to aid clarity. Preferably, the distal end of the medicament container **222** has a septum assembly **228** that is spaced apart from the valve assembly **212** prior to actuation of the device **222**, as best shown in FIG. 47.

[0164] For manufacturing purposes, using one size for a medicament container is often desirable, even if multiple fill volumes or dosages are contemplated for use with the container. In such cases, when medicament containers are filled, the differing fill volumes result in different positions of the stopper. To accommodate such different stopper positions, as well as accommodate manufacturing differences of the stoppers, aspects of the present invention include a bespoke or custom spacer **226** disposed in a proximal end of the container **222**, proximal to the stopper **224**. In other words, the bespoke spacer **226** provides an option that allows dispensing of a range of manufacturer-set pre-defined fill volumes by selection of different spacers **226**, and reduces or eliminates the need for assembly configuration operations. The size of the spacer **226** can be employed to account for under-filled volumes of the container **222**, and provide a consistent bearing surface at the proximal end of the container.

[0165] The spacer **226** is selected from a plurality of different size spacers **226** to occupy space from a proximal end of the stopper **224** to a proximal end of the container **222**. According to one embodiment, as shown in FIGS. 45-47, the spacer **226** is selected to be substantially flush with the proximal end of the container **222**. Additionally, according to one embodiment, the spacer **226** has a “top hat” shape, which includes a central column **230** and a distal flange **232**, as best shown in FIG. 45.

[0166] Returning to FIGS. 44-47, the system **200** also includes a drive assembly **234** for displacing the container **222** distally to establish the fluid connection between the container **222** and the patient needle **215**, as well as dispensing the medicament from the container **222**. In more detail, the drive assembly **234** includes an inner spring **236** disposed within a central plunger **238**, an outer plunger **240**, an outer spring **242** disposed between the central plunger **238** and the outer plunger **240**, a telescoping member **244**, and a release gate **246**.

[0167] Preferably, the inner spring **236** has a greater spring constant than the outer spring **242**, and is therefore, stronger or stiffer than the outer spring **242**. The inner spring **236** is disposed inside the central plunger **238**, and pushes between a spring flange **248** in the lower housing (best shown in FIG. 46) and the central plunger **238**, which bears directly on the proximal end of the spacer **226** subsequent to device activation. The outer spring **242** is disposed inside outer plunger **240**, and pushes between a proximal external flange **250** of the central plunger **238** and a distal internal flange **252** of the outer plunger **240**. Thus, the inner and outer springs **236** and **242** are nested, and can provide a more compact drive assembly (and thus, a more compact system **200**) than employing a single spring.

[0168] According to one aspect, the inner spring **236** acts only to displace the container **222** to establish the fluid connection with the patient needle **215**, and the outer spring **242** acts only to subsequently dispense the medicament from the container **222**. According to another aspect, the inner spring **236** acts to displace the container **222** to establish the fluid connection with the patient needle **215**, and also acts to begin dispensing the medicament from the container **222**, and the outer spring **242** acts to complete dispensing the medicament. In a further aspect, the inner spring **236**

causes the initial piercing of the container **222** with the outer spring **242** completing the piercing and dispensing of the medicament from the container **222**.

[0169] As shown in FIGS. **44-47**, and as subsequently described in greater detail, the outer plunger **240** includes a pair of proximal flanges or feet **254** that each have a slanted surface that interacts with a corresponding slanted surface (or surfaces) on the release gate to retain and subsequently release the power module subsequent to actuation of the device **200**.

[0170] As best shown in FIGS. **46** and **47**, as initially assembled, the container **222** is disposed in clearance from the drive assembly **234** and the valve assembly **212**. A lateral flange **256** on the needle actuator **220** axially retains the medicament container **222**, and the needle actuator **220** prevents the release gate **246** from displacing laterally. According to one embodiment, a spring (not shown) biases the needle actuator **220** distally, but the actuation button **210** (and/or its associated assembly) prevents distal displacement of the needle actuator **220** prior to actuation of the device **200**. A status bar **258** is disposed on the needle actuator **220**, and has a top surface that is visible through the status view port **208**. According to one embodiment, the top surface of the status bar has a plurality of colors or patterns, and when the device is in a pre-actuated state, a first color or pattern, such as yellow, is visible through the status view port **208**.

[0171] FIGS. **48-52** are top views of the system **200** illustrating the operation of events at and subsequent to actuation of the system **200**. In FIG. **47**, a user slides the actuation button **210** proximally and then displaces the button **210** vertically into the housing **202**, thereby freeing the needle actuator **220** to displace distally under the influence of the spring (omitted for clarity). As shown in FIG. **49**, as the needle actuator displaces distally, tracks **260** on the needle actuator **220** interact with lateral bosses **262** on the needle arm **216** to insert the patient needle **215**. Preferably at this stage, the proximal end of the needle actuator **220** has not yet cleared the release gate **246**, and thus, the drive assembly **234** has not yet been released. But the lateral flange **256** has displaced distally and therefore, the container **222** is unrestrained.

[0172] Subsequently, as shown in FIGS. **50** and **51**, with continued distal displacement, the proximal end of the needle actuator **220** clears the release gate **246** (thereby releasing the drive assembly **234**). The needle actuator **220** comes to temporarily rest against a feature on a rotatable release flipper **264**, driving the release flipper **264** against an outrigger **266** (best shown in FIGS. **44** and **59**) of the telescoping member **244**. The needle actuator **220** remains in this position until the medicament has been dispensed. In this position, preferably, a second color or pattern of the status bar **258**, such as green, is visible through the status view port **208**.

[0173] At this stage, the force of the springs **236** and **242** and the interaction of the angled surfaces of the proximal flanges or feet **254** with the corresponding angled surface (or surfaces) on the release gate **246** causes the release gate **246** to displace laterally, thereby freeing the outer plunger **240** from restraining interaction with the release gate **246**. Up to this point, the outer plunger **240** has been restraining the central plunger **238**.

[0174] Referring to FIGS. **52** and **53** (the inner spring **236** is omitted from FIG. **52** for clarity), the stiff inner spring **236** distally drives central plunger **238** to contact the spacer **226**. Because the medicament container **222** is filled with a substantially incompressible fluid, the continued distal displacement of the central plunger **238** distally displaces the spacer **226**, the stopper **224**, and the container **222** relative to the housing **202**. This distal displacement causes the septum assembly **228** to be pierced by the valve assembly **212**, establishing fluid communication between the container **222** and the patient needle **215**. The central plunger **238** travels distally until its proximal external flange **250** (best shown in FIG. **59**) contacts a flange on the lower housing **206**, thereby limiting the “piercing travel.” Preferably, another flange on the lower housing **206** and/or the lateral flange **256** of the needle actuator **220** limits distal travel of the container **222**.

[0175] Subsequently, because the inner spring **236** can no longer distally displace the central plunger **238**, the lighter outer spring **242** distally displaces the outer plunger **240** relative to the central plunger **238** to contact the distal flange **232** of the spacer **226**, as shown in FIGS. **54** and **55**.

As subsequently described in greater detail, preferably, the contact between the outer plunger **240** and the spacer **226** is damped to minimize the impact force. Further expansion of the outer spring **242** distally displaces the outer plunger **240** to dispense the medicament.

[0176] As shown in FIGS. **56** and **57**, as the outer spring **242** continues to expand and distally displace the outer plunger **240**, upon a predetermined distal displacement of the outer plunger **240** relative to the telescoping member **244**, an external feature or flange **268** of the outer plunger **240** interacts with an internal distal feature or flange **270** of the telescoping member **244** to “pick up” the telescoping member **244**. This ensures that further distal displacement of the outer plunger **240** causes corresponding distal displacement of the telescoping member **244**. This paired distal displacement continues until the end of the medicament dispensing.

[0177] As previously noted, the outrigger **266** is disposed on the telescoping member **244**. The axial length of the outrigger and the distal travel of the telescoping member **144** controls the timing of the disengagement of the outrigger **266** with the release flipper **264**. As shown in FIGS. **58** and **59**, at the end of medicament dispensing, the proximal end of the outrigger **266** bypasses the release flipper **264**. This allows the release flipper **264** to rotate out of engagement with the needle actuator **220** (FIG. **60**), and allows the needle actuator **220** to continue its distal displacement and withdraw the patient needle **215** (FIG. **61**). At this stage, another color or pattern of the status bar **258**, such as red, is visible through the status view port **208**, signifying that the device **200** has completed operation.

[0178] As previously noted, the contact between the outer plunger **240** and the spacer **226**, as illustrated in FIGS. **62** and **63**, is preferably damped to minimize the impact force. The highest level of energy dissipation is desirable for under-filled syringes containing viscous fluid, as the outer spring **242** will be stiffer to provide desired dispense rates. The lowest level of energy dissipation is desirable for maximum-filled syringes containing low-viscosity fluid, as the outer spring can be less stiff to provide desired dispense rates. Various methods can be employed to adjust damping levels, such as air damping, or closed-cell foam damping.

[0179] As another method of damping the impact force, FIG. **64** illustrates an embodiment of a spacer **226** in which one or more axial interface ribs **272** are circumferentially arrayed about the central column **230** of the spacer **226**. In this embodiment, the outer plunger **240** must drive past the interference ribs **272**, which provide frictional resistance to the distal displacement of the outer plunger **240** relative to the spacer **226**. The frictional force created by the interference between interference ribs **272** and the outer plunger **240** is independent of plunger speed. Preferably, the frictional force does not exceed the minimum dispense spring load, to avoid stalling weaker springs. The interference can be tuned to give the desired level of frictional resistance. For different fluid viscosities, there can be different sizing (axial and/or radial) of the interference ribs **272**. This could mean a bespoke or custom spacer for each viscosity and fill-level combination, or, depending on the number of springs required for a viscosity range, there can be a number of tuned positions, whereby the spacer can be set to a particular position for a particular modular spring (the position have had the interference/damping tuned for that particular spring load/viscosity scenario).

[0180] Referring to FIGS. **65A-69**, an actuator button arrangement **280** for actuating the system **10** according to one aspect of the present invention is shown. The actuator button arrangement **280** includes the actuator button **26**, a button spring **284**, and a needle actuator body **286**. The needle actuator body **286** may be similar to the needle actuator bodies **96**, **220** discussed above and configured to move within the housing **20** to transition the needle shuttle **102** or needle **28** between retracted and extended positions. As shown in FIG. **69**, the actuator button **26** includes a user interface portion **288** for interacting with a user. Preferably, the user interface portion **288** is about 22 mm long and about 10 mm wide, although other suitable dimensions may be utilized. The actuator button **26** includes two pairs of lockout arms **290**, **292** that interact with button contacting surfaces **294**, **296** on the needle actuator body **286** prior to device actuation to prevent the needle actuator body **286** from rocking upward. As shown in FIG. **65H**, an overlap between the needle

actuator body **286** and the housing **20** prevents premature actuation. Referring to FIG. **66**, the button spring **284** includes a first bearing surface **298** and a second bearing surface **300** spaced from the first bearing surface **298**, and a cantilevered central spring arm **302** surrounded by a pair of outer arms **304** that are joined by the first bearing surface **298**.

[0181] The actuation button arrangement **280** is configured to provide one or more of the following features, which are discussed in more detail below: one-way axial displacement or sliding of the actuator button **26**; transverse movement (raised and depressed positions) of the actuator button **26** where the actuator button **26** remains depressed during the use position of the needle actuator body **286**; and lockout of the actuator button **26** in the post-use position of the needle actuator body **286** such that the button **26** is in the raised position and cannot be depressed by a user.

[0182] To actuate the system **10** using the actuator button **26**, the user first slides the user interface portion **288** in a first axial direction, shown as being to the right in FIGS. **65G** and **65H**. The user may be required to slide the user interface portion **288** about 10 mm or about 8 mm, although other suitable distances may be utilized. Moving the actuator button **26** axially moves the lockout arms **290**, **292** to clear the button contact surfaces **294**, **296** on the needle actuator body **286** to allow movement of the actuator button **26** from the raised position to the depressed position.

[0183] As the user distally slides the user interface portion **288**, the central spring arm **302** of the button spring **284** rides over a spring arm **306** bearing surface on the housing **20** while the first and second bearing surfaces **298**, **300** engage first and second bearing ramps **308**, **310** on the housing **20**. The forces on the button spring **284** are balanced through the engagement with the spring arm bearing surface **306** and the first and second bearing ramps **308**, **310** to provide a smooth axial displacement or sliding of the actuator button **26**.

[0184] As the actuator button **26** and the button spring **284** reach the end of their axial sliding travel, the central spring arm **302** and the first bearing surface **298** pass the end of a respective stops **312**, **314** to prevent the actuator button **26** from sliding backward to its original position, as shown in FIG. **65H**. Further, when the actuator button **26** and the button spring **284** reach the end of their axial sliding travel, the user engages the user interface portion **288** to move the actuator button **26** downward to its depressed position. The actuator button **26** may be depressed about 2 mm and the minimum force required to depress the actuator button **26** is about 3 N, and most preferably, about 2.8 N, although other suitable distances and minimum forces may be utilized.

[0185] As the user depresses the user interface portion **288**, shown in FIGS. **65A** and **65B**, the actuator button **26** rotates the needle actuator body **286** to release the needle actuator body **286** thereby allowing the needle actuator body **286** to move from the pre-use position to the use position. As shown in FIG. **65B**, as the needle actuator body **286** travels to the use position, the lockout arms **290**, **292** run along the underside of the button contact surfaces **294**, **296** to prevent the actuator button **26** springing upward. After the medicament has been delivered and as the needle actuator body **286** is transitioning from the use position to the post-use position, shown in FIG. **65C**, the lockout arms **290**, **292** are disengaged from the button contact surfaces **294**, **296** allowing the actuator button **26** to spring back up under the influence of the button spring **284**. Once the needle actuator body **286** fully transitions to the post-use position, shown in FIG. **65D**, the actuator button **26** has finished moving from the depressed position to the raised position due to the biasing force of the button spring **284**. When the needle actuator body **286** is in the post-use position, a spring arm **316** on the needle actuator body **286** engages the actuator button **26** to prevent the actuator button **26** from moving to the depressed position while axial movement is still restricted by the engagement of the spring arm **302** with the stops **312**, **314**. Thus, the actuator button **26** is locked after delivery of the medicament is complete to provide a clear indication between a used system and an unused system.

[0186] Furthermore, if the user holds down the actuator button **26** during dispensing of the medicament, proper dosing and needle retraction will still complete, but the actuator button **26** will not spring back up to the raised position until the button **26** is released.

[0187] In one aspect, the button spring **284** is made of plastic. The button spring **284** may also be a pressed metal spring could be used instead, although any other suitable material may be utilized.

[0188] Referring to FIGS. **68A-68G**, rather than providing a separate actuator button **26** and button spring **284**, the spring may be provided integrally with the button **26**. More specifically, an actuator button **320** according to a further aspect of the present invention includes an integral spring arm **322**. The actuator button **320** also includes lockout arms **324**, retention arms **326**, and a rear pivot **328**. As shown in FIGS. **68D** and **68E**, the spring arm **322** engages prongs **330** in the top portion **22** of the housing **20**. During transition of the system **10** from the pre-use position to the use position, the spring arm **322** slides past a detent of the prongs **330** providing an axial spring force. The end of the spring arm **322** engages a portion of the top portion **22** of the housing **20** to provide the vertical spring force as the spring arm **322** deflects. The actuator button **320** is configured to a fluid motion between the sliding and depression movements of the button **320** even though two separate motions are occurring, which is similar to the operation of the button **26** discussed above. During transition between the pre-use position and the use position, the button **320** pivots about the rear pivot **328** with the retention arm **326** engaging a portion of the needle actuator body **286** thereby maintaining a depressed position of the button **320** until the end-of-dose position is reached in a similar manner as actuator button **26**. The lockout arms **324** deflect inwards and engages a portion of the needle actuator body **286** as the needle actuator body **286** moves to the end-of-dose position thereby preventing further movement of the actuator button **320** in a similar manner as the actuator button **26** discussed above.

[0189] Aspects of the present invention provide improvements over previous button designs. For example, the actuation button arrangement **280** provides multiple surfaces to hold the needle actuator body **286** in place against a needle actuator spring **106** prior to actuation, thereby reducing the likelihood of premature actuation during a drop impact. The actuation button arrangement **280** physically prevents the needle actuator body **286** from moving prior to actuation by holding it in a tilted (locked) state in such a way that the surfaces have no room to separate and pre-activate.

[0190] In addition, button slide forces of the actuation button arrangement **280** are controlled more precisely by utilizing a flexing arm rather than using a simple bump detent. This permits longer sliding strokes of the button **26** with better force control, resulting in a more ergonomically effective design. Further, the actuation button arrangement **280** causes the button **26** to pop back out at the end of injection, giving the user an additional visual, audible, and tactile indication that the medicament delivery is completed.

[0191] According to one aspect, the fluid delivery volume of the system **10** is determined by the end position of a plunger relative to a point inside the housing regardless of actual fill volume, container inner diameter, and stopper starting position and length. The dosing accuracy variability can be significant because the tolerances of the factors above can be quite large. Aspects of the present invention allow for the elimination of some or all of these tolerances from the dosing equation, resulting in a more precise and less variable injection volume of medicament.

[0192] Referring to FIGS. **70A-70G**, a spacer assembly **400** for use in connection with a drive assembly according to one aspect of the present invention is shown.

[0193] Elements in a chain of tolerances in the stopper spacer assembly **400** include a thickness (A) of a flange **402** of an inner plunger **404**, an internal length (B) of an outer plunger **406** between an internal proximal end **408** and an internal shoulder **410**, and an initial offset distance (C.sub.1) between the inner plunger flange **402** and the internal proximal end **408** of the outer plunger. This initial offset distance (C.sub.1) is preferably greater than a gap distance (C.sub.2) between outer plunger **406** and the proximal end of the medicament barrel **412**. The chain of tolerances in the stopper spacer assembly **400** also includes the internal barrel diameter (D). Once assembled, the stopper spacer **414** and the outer plunger **406** are unique for a given medicament volume.

[0194] FIGS. **70B-70G** illustrate operation of the stopper spacer assembly **400**. As shown in FIG. **70B**, when the system is actuated, the both inner and outer plungers **404** and **406** are released. An

outer spring **416** pushes the outer plunger **406** into the barrel **412**, compressing damping material **418**, and an inner spring **420**. The stopper **422** does not yet move relative to the barrel **412** due to the fluid column of medicament.

[0195] Next, as shown in FIGS. **70C**, the outer spring **416** distally displaces the outer plunger **406** and the barrel **412** to open a valve (not shown) at the distal end of the barrel **412** that establishes fluid communication with the needle (not shown). Due to the incompressibility of the liquid medicament, the stopper **422** cannot displace relative to the barrel **412** until the valve is opened and the fluid path to the patient needle is established.

[0196] Subsequently, as shown in FIGS. **70D** and **70E**, the inner spring **420** displaces the inner plunger **404**, the stopper spacer **414**, and the stopper **422**, to dispense the fluid.

[0197] FIG. **70F** illustrates the end of medicament delivery when the proximal flange **402** of the inner plunger **404** contacts the internal shoulder **410** of the outer plunger **406**, thereby ceasing displacement of the inner plunger **404** (and the stopper spacer **414** and stopper **422**) relative to the medicament barrel **212** and stopping the flow of medicament.

[0198] According to one aspect, as shown in FIG. **70G**, the cessation of displacement of the inner plunger **404** relative to the medicament barrel **412** triggers an end-of-dose indicator for the system.

[0199] Referring to FIGS. **71** and **72**, a collapsible spacer assembly **430** includes a forward spacer portion **432** secured to a stopper **434**, an inner plunger **436**, a rear spacer portion **438**, and a rotating shuttle **440**. The inner plunger **436** can translate relative to the forward spacer portion **432**, but not rotate relative thereto. Similarly, the rear spacer portion **438** can also move axially relative to the forward spacer portion **432**, but not rotate relative to the forward spacer portion **432**. As subsequently described in greater detail, the rotating shuttle **440** first rotates, and subsequently translates.

[0200] According to one aspect, forward spacer portion **432** is fixedly secured to the stopper **434**. One skilled in the art will understand that many methods can be employed to secure the forward spacer portion **432** to the stopper **434**, for example, adhesive, mechanical fasteners, or any other suitable arrangement. Preferably, the forward spacer portion **432** includes threads that engage mating threads in the stopper **434**.

[0201] When the stopper spacer assembly **430** is screwed into the stopper **434**, an axial load is applied through access openings **442** in the rear spacer portion **438**. This force can be used to push the stopper **434** forward, applying pressure to the fluid medicament. This pressure causes the front (distal) face of the stopper **434** to deflect and press proximally, pushing back on the rear spacer portion **438** and rotating the rotating shuttle into its “as assembled” condition. In other words, when a medicament barrel is filled with medicament and the system's plunger is applying axial force to the medicament via the spacer assembly **430**, the distal face of the stopper **434** is deformed by the pressure of the medicament. During medicament delivery, pressure is applied by a drive assembly (via the plunger) to the rear spacer portion **438**, which in turn applies a rotational torque to the rotating shuttle **440** via helical faces **444** of the rear spacer portion **438**. But the stopper deformation from the medicament provides a rearward or proximal force on the inner plunger **436**, which prevents rotation of the rotating shuttle **440**.

[0202] According to one aspect, an axial reaction load on the inner plunger **436** can be increased by increasing the length of the inner plunger **436**.

[0203] Once the medicament delivery is complete, as shown in FIG. **73**, the pressure on the stopper **434** decreases, thereby permitting the distal end of the inner plunger **436** to displace distally. This distal displacement permits the rotating shuttle **440** to rotate. The continued axial force applied by the drive assembly rotates and distally displaces the rotating shuttle **440** due to interaction of the helical faces **444** in the rear spacer portion **438** with corresponding cam-faced arms **446** of the rotating shuttle **440**. According to one aspect, this final movement of the rotating shuttle **440** causes the drive assembly to trigger needle retraction.

[0204] Referring to FIGS. **74** and **75**, a restriction member **452** according to one aspect of the

present invention is disposed with the drive assembly. The restriction member **452** governs the timing of the final displacement of the needle actuator bodies **96, 220** subsequent to the completion of the medicament dose. Instead of rotating about a fixed post, the restriction member **452** floats freely. Once a plunger displaces sufficiently distally for a gap to align with the restriction member **452** (as shown in FIGS. **74** and **75**), the restriction member **452** displaces laterally into the gap because of the force of the spring on the needle actuator **96, 220** and the angled face **454** on the rear of the arm of the restriction member **174** that engages the needle actuator body (best shown in FIG. **75**). Once the restriction member no longer retains the needle actuator body **96, 220**, the needle actuator body **96, 220** is free to complete the axial movement to the post-use position. Further, as shown in FIG. **75**, the restriction member **452** is biased onto the rear of the barrel portion of the container **14**, which minimizes the tolerance chain of the various components and improves dose accuracy.

[0205] Referring to FIGS. **76-78**, a spacer assembly **460** according to a further aspect of the present invention is shown. The spacer assembly **460** shown in FIGS. **76-78** allows for the removal of the effect of manufacturing tolerance build up through adjustment of the spacer assembly thereby allowing each system to inject the same amount of medicament.

[0206] As shown in FIG. **77**, the spacer assembly **460** includes a stopper **462** and a stopper spacer **464**. The stopper spacer **464** includes a fixed spacer piece or fixed spacer **466** that is fixedly connected with the stopper **462**, and an adjustable spacer piece or adjustable spacer **468** that is rotationally displaceable in one direction relative to the fixed spacer **466**.

[0207] One skilled in the art will understand that many methods can be employed to secure the fixed spacer **466** to the stopper **462**, for example, adhesive, mechanical fasteners, or any other suitable arrangement. Preferably, the fixed spacer **466** includes one or more external threads that engage one or more mating threads in the stopper **462**. According to one aspect, the adjustable spacer **468** has a distal stem with an external thread **470**. The distal stem thread **470** engages an internal thread **472** in the fixed spacer **466** (best shown in FIG. **78**) to rotationally control axial displacement of the adjustable spacer **468** relative to the fixed spacer **466**.

[0208] As shown in FIGS. **76** and **77**, the fixed spacer **466** includes radially spaced detents **474** and the adjustable spacer **468** includes a spring detent arm **476**, the free end of which engages a selected one of the detents **474** to prevent rotation and axial displacement of the adjustable spacer **468** toward the fixed spacer **466**. The free end of the spring detent arm **476** is shaped to pass over the detents **474** in one direction, thereby permitting rotation and proximal axial displacement of the adjustable spacer **468** away from the fixed spacer **466**.

[0209] Despite variations in the dimensions of stoppers and containers, the adjustable spacer **468** can be adjusted relative to the fixed spacer **466** to provide a consistent axial length of the stopper assembly **460**.

[0210] As shown in FIG. **78**, once the container is filled, an axial load, such as a load that would be encountered when installed in the system **10, 200**, can be applied to the adjustable spacer **468** (and thus, the fixed spacer **466** and the stopper **462**). Once the axial load is applied, the adjustable spacer **468** can be proximally backed out to ensure a consistent gap **478** between the proximal end of a medicament barrel **480** and the proximal face of the adjustable spacer **468**, thereby accounting for variations in the medicament barrel glass and the compressibility of any entrapped air. In other words, the spacer assembly **460** allows the adjustable spacer **468** to have a predetermined set position relative to the container **14** independent of the variables of the container **14** and stopper length. Accordingly, the start position of the spacer assembly **460** is a predetermined distance from the container **14** and the end position of the spacer assembly **460** is also a predetermined distance from the container **14** such that the travel of the stopper **462** is defined by the effective length of the plungers **52, 54** of the drive assembly **12**.

[0211] Referring to FIGS. **79** and **80**, a base column **482** and a cap **484** of an automatically adjusting spacer **486** according to one aspect of the present invention is shown. The base column

482 includes a base portion **488** and an axially extending column **490**. According to one embodiment, the base column **482** includes a plurality of columnar protrusions **491** that each have a plurality of ratchet teeth **492** disposed on a proximal portion thereof. A locking barb **493** is disposed at the proximal end of each of the plurality of ratchet teeth **492**. The cap **484** is hollow, and a distal end of the cap **484** includes one or more axial springs **494**. According to one aspect, the axial springs **494** are bent, cantilevered arms formed during molding of the cap **484**. According to another aspect, a separate biasing member, such as a compression spring can be employed in the automatically adjusting spacer **486**. When assembled with the base column **482**, the springs **494** engage the base portion **488** and maintain an initial spacing between the base column **482** and the cap **484**. According to one aspect, the springs **494** are omitted. The cap **484** also includes a plurality of flexible cantilevered arms or tabs **496**, which each have a free proximal portion with a plurality internal of ratchet teeth **497**. The proximal end of each flexible tab **496** includes a foot **498**.

[0212] FIG. **81B** illustrates the cap of the automatically adjusting spacer deployed within a proximal recess of a stopper **494** at a proximal portion of a medicament barrel. The base column **482** is assembled into the hollow cap **484** with the base portion **482** engaging the stopper **494** and the feet **498** disposed outside the proximal end of the barrel.

[0213] In operation, as shown in FIGS. **81A** and **81B**, the cap **484** displaces distally relative to the base column **482** (as well as the stopper **494** and the barrel) until the proximal end of the cap **484** is flush with the end of the medicament barrel. This action causes the feet **498** to engage the internal surface of the barrel and displace radially inward, thereby forcing the ratchet teeth **492** into locking engagement with the ratchet teeth **497**. The locking barb **493**, the engagement of the ratchet teeth **492** and **497**, and the engagement of the feet **498** with the internal surface of the barrel prevents the displacement of the cap **484** relative to the base column **482**. Thus, the automatically adjusting spacer **486** can accommodate differences in stoppers, barrel diameters, and medicament fill volumes, to automatically provide a bearing surface flush the proximal end of the medicament barrel.

[0214] One aspect of the present invention is a spacer assembly **486** that is situated against the stopper in the container within the system. The spacer design is such that its effective length can be adjusted in order to allow the dispensing of a precise quantity of medicament. The length adjustment is intended to compensate for manufacturing tolerances within the container, the fill volume, and especially the stopper length, which can add up to $\frac{1}{3}$ of the variability in a delivered dose using a non-adjustable spacer. The spacer length can be adjusted through several techniques, depending on the specific aspect. The spacer length can be self adjusting based on its location to the back of the container, it can be adjusted by assembly equipment at the time of final assembly of the primary container into the subassembly, and it can be made an integral part of the stopper and adjusted as a subassembly prior to filling. The adjustable spacer **486** allows a more precise volume of fluid to be injected compared to a non-adjustable stopper.

[0215] Referring to FIGS. **82-87**, a drive assembly **500** for a drug delivery system according to one aspect of the present invention is shown. The drive assembly **500** includes an actuation button **506**, a container **508**, a needle actuator assembly **510**, an actuation release or flipper **512**, a lead screw **514**, and a plunger **516**. The lead screw includes a drum portion **518** with external radially-protruding vanes **520**, and, as best shown in FIGS. **84** and **85** and subsequently described in greater detail, a screw thread portion **522**. Prior to activation, as best shown in FIGS. **83** and **86** one end **513** of the actuation release **512** engages one of the vanes **520** to prevent rotation of the lead screw **514**.

[0216] According to one aspect, as shown in FIGS. **84-86**, the screw thread portion **522** of the lead screw **514** engages internal threads of a nut **524** connected with the plunger **516**. According to another aspect, the nut and its internal threads are integrally formed with the plunger as a unitary structure. Additionally, a constant force spring **526** is received within the drum portion **518** and biases the lead screw **514** in a rotational direction. According to one aspect, the spring **526** is

secured to the base cover **504**. According to another aspect, as shown in FIGS. **84-86**, a drive assembly housing **528** is disposed within the system and the spring **526** is secured to the power pack housing **528**.

[0217] Unlike a helical spring, such as a compression spring, which has a force profile proportional to its displacement, the constant force spring **526** and the like maintain a relatively flat or even force profile over a long working length. The even force profile advantageously provides an injection force that is proportional to the spring force. This will provide a flat or even injection force, and thus, a substantially constant injection rate for the medicament. Although the spring **526** is illustrated in FIG. **86** as having only two turns of material, one skilled in the art will appreciate that fewer or greater numbers of turns can be employed. Preferably, an assembler winds the spring **526** when the drive assembly **500** is assembled, and the spring **526** is stored in the wound position until the time of actuation.

[0218] Upon actuation of the system, the needle actuator assembly **510** is released to axially displace (to the right in FIGS. **82-85**) from the pre-use position to the post-use position under the influence of a biasing member **530** (best shown in FIG. **83**). During this displacement, the needle actuator assembly **510** bears against a second end **532** of the actuation release **512** and rotates the release **512** counter-clockwise, as shown in FIG. **87**. This counter-clockwise rotation of the actuation release **512** frees the first end **513** thereof from engagement with the vane **520**.

Subsequent to the disengagement of the first end **513** from the vane **520**, the spring **526** unwinds and drives rotation of the lead screw **514**, which, in combination with the nut **524**, advances the plunger **514** to dispense the medicament.

[0219] As the lead screw **514** is rotating, the rotation of the drum portion **518** and the vanes **520** is visible through a window **534** in the housing. This window **534** indicates progress of the screw in a way that is much more apparent than viewing the linear movement of the stopper **536** in the container **508**. In fact, this rotational movement is many times more sensitive than the linear movement. One skilled in the art will appreciate that the exact amount of advantage or increase depends on the pitch of screw thread portion **522** of the lead screw **514**, the diameter of the drum portion **518**, and number of vanes **520** on the drum portion **518**.

[0220] Referring to FIGS. **88-93**, a drive assembly **600** for a drug delivery system according to a further aspect of the present invention is shown. The drive assembly **600** acts to store a spring's mechanical energy and to activate it when triggered. The drive assembly **600** includes a medicament barrel **601**, a stopper **602** slidably disposed in the barrel **601**, a first valve plunger **603**, a second valve plunger **604**, a first revolve nut **605**, and a second revolve nut **606**. The drive assembly **600** also includes a rotary indicator **607**, a locking element **608**, a constant force spring **609** disposed within the rotary indicator **607**, and an actuation release or flipper **610**. The drive assembly **600** is at least partially disposed within a housing **611** that can be assembled into a drug delivery system.

[0221] The constant force spring **609** is contained between the housing **611** and the rotary indicator **607** within a drum portion **616** of the rotary indicator **607**. The drive assembly's inactive state is such that energy is applied by uncoiling the spring **609** and harnessing this energy geometrically with the housing **611**, rotary indicator **607**, and actuation release **610**. When the drive assembly **600** is deactivated, the spring recoils and translates the mechanical energy into rotational motion of the rotary indicator.

[0222] The telescoping multi-part plunger is oriented along a force axis between the medicament barrel **601** and the rotary indicator **607**. The rotary indicator **607** features a threaded shaft **618**. According to one aspect, the threads are dual lead, and are either square or rectangular in nature. The multi-part telescoping plunger includes a two-part threaded nut (first revolve nut **605** and second revolve nut **606**) and a two-part plunger (first valve plunger **603** and second valve plunger **604**). The second revolve nut **606** is a threaded shaft that mates with the rotary indicator **607** and first revolve nut **605** and features matching threads on its inner and outer surfaces (internal and

external threads, respectively) to mate with them. The second revolve nut **606** also has a circular collar **620** (best shown in FIG. **92**) on its proximal end that bottoms down on the second valve plunger **604**. The second revolve nut **606** is free to spin along the force axis. The first revolve nut **605** is also a threaded shaft that features threads on its inner diameter corresponding to the external threads of the second revolve nut **606** to mate with the second revolve nut **606**.

[0223] According to one aspect, on one end, the first revolve nut **605** has a hexagonal collar that press fits on the first valve plunger **603** to fixedly connect the first valve plunger **603** with the first revolve nut **605**. In the drive assembly **600**, the first revolve nut is not free to rotate and will only translate when the power module subassembly is actuated.

[0224] The second valve plunger **604** is a hollow cylindrical component with a small collar **622** on its distal end, a large collar **624** on its proximal end, and an extended L-shaped arm **626** (best shown in FIG. **93**) protruding from the large proximal collar **624**. According to one embodiment, the small collar **622** is discontinuous and features four leaf cantilevered arms or leaf springs **623** that allow the collar to bend and mate with the first valve plunger **603**. The inner surface of the second valve plunger **604** has an undercut through its length terminating at its proximal end a radially inward protruding shelf **628** of the large collar **624**. The shelf **628** engages the second revolve nut **606** within the telescoping assembly.

[0225] The first valve plunger **603** attaches to the stopper **602** and is also a hollow cylindrical component that mates with the second valve plunger **604**. More specifically, the first valve plunger **603** features a cylindrical protrusion **630** on its distal end to mate with the stopper **602**. According to one aspect, as best shown in FIG. **89**, four thru slots **632** are disposed on the proximal quadrants of the first valve plunger **603** to mate with the leaf springs or arms **623** and small collar portion **622** of the second valve plunger **604**. Both the first and second valve plungers **603** and **604** are free to slide.

[0226] Telescoping is achieved when the constant force spring **609** recoils and the rotary indicator **607** starts spinning. The threaded attachment between the rotary indicator **607** and the second revolve nut **606** causes second revolve nut **606** to rotate. But because the second revolve nut **606** is threaded to the first revolve nut **605**, which cannot rotate and experiences resistance to distal translation due to the pressure caused by medicament in the barrel **601**, the second revolve nut **606** will displace proximally and bottom out on the second valve plunger's radially inward protruding shelf **628**. The second valve plunger **604** is prevented from displacing proximally by the housing **611**. Subsequently, and with continued rotation of the rotary indicator **607**, because the second revolve nut **606** is threaded with the first revolve nut **605** (which cannot rotate) the first revolve nut **605** translates distally to push the first valve plunger **603** (and the stopper **602**) to dispense medicament from the barrel **601**.

[0227] The first valve plunger **603** displaces distally relative to the second valve plunger **604** until the small collar sections **622** (respectively disposed on the distal ends of the leaf springs or arms **623** of the second valve plunger **604**) engage the corresponding proximal ends of the slots **632** of the first valve plunger **603**. This locks the relative position of the first and second valve plungers **603** and **604**, with continued rotation of the rotary indicator **607**, both valve plungers translate distally while also pushing the second revolve nut along (because of its proximal engagement with the shelf **624**).

[0228] The initial and final positions of the telescoping plunger, and thus the medicament dose, are controlled by the rectangular thread form of the threaded shaft **618** of the rotary indicator **607**, a threaded shaft on the drum portion **616** of the rotary indicator **607**, and a stepped pin that acts as the locking element **608**. According to one aspect, threaded shaft on the drum portion **616** of the rotary indicator **607** is single lead, and because the rest of the components in the telescoping chain have dual lead threads, the axial travel of the other threaded components is twice the axial travel of the lock **608** relative to the rotary indicator.

[0229] According to one embodiment, the lock **608** is cylindrical and features a domed tip on one

end and a cylindrical collar on the other. The threads on the exterior of the rotary indicator's drum portion **616** along with a slot and undercut **636** at the bottom of the housing **611** captures the lock **608** in place, allowing it to slide parallel to the force axis. Thus, as the spring **609** is released and the rotary indicator **607** turns, the lock **608** translates as well and creates a positive stop when the distal end of the thread on the exterior of the rotary indicator's drum portion **616** is reached.

[0230] One benefit of aspects of the drive assembly **600** include the use of a constant force spring **609**, the mechanical energy of which is converted into substantially constant linear force to the medicament in the barrel **601**. In turn, this creates a uniform medicament delivery rate. Another benefit is that employing the telescoping plunger driven by a thread form, the drive assembly can create in-line space savings of up to 0.75 inches compared to other plunger designs. Additionally, the drive assembly provides a controlled medicament dose through an initial and final mechanical constraint within the same component.

[0231] As previously noted, other drug delivery systems utilize a compressed coil spring, which exerts a maximum force at actuation that eventually decreases as the spring expands. A decreasing force at the plunger translates into variable medicament delivery time and medicament exit pressure. By using a constant force spring, the force exerted on the plunger is constant from the beginning to the end of the dosage. In addition, the distance a coil spring has to travel in addition to the length of a static plunger that needs translate inside the drug container can create a long assembly. In contrast, in embodiments of the present invention, the constant force spring is contained radially and does not require any additional space before or after activation. Furthermore, the aspects of the telescoping plunger allow that the plunger length of the can be significantly reduced in comparison to the length of a static plunger.

[0232] Previous drug delivery systems have variable dose accuracy performance because the mechanical components enabling the drug delivery create a geometric dependence by bottoming down on the container, which cannot be fabricated with tight tolerances. Some embodiments of the present invention create a control to the start and end times of the translating plunger via a thread form in the rotary indicator and the use of the constant force spring.

[0233] The drive assembly creates a space saving geometry in addition to well-controlled time, volume and pressure for the drug delivery device, which translates to a more attractively compact and precise drug delivery device.

[0234] Some aspects of the drive assembly implement three rotating threaded shafts to create a linear space savings of about 0.75 inch. In other aspects, the same concept can be employed using two rotating threaded shafts and result in a space savings of about 0.5 inch. Some aspects of the present invention convert the rotational energy of a constant force spring to a translational force motion of a plunger.

[0235] Referring to FIGS. **94-100**, a spacer assembly **660** according to a further aspect of the present invention is shown. The spacer assembly **660** is similar to the spacer assembly **460** discussed above and shown in FIGS. **76-78** and operates in a similar manner to achieve similar advantages. The spacer assembly **660** includes a fixed spacer **666** and an adjustable spacer **668**. The fixed spacer **666** is configured to be received by the stopper **462** with lugs **670** engaging the stopper **462** to secure the fixed spacer **666** within the stopper **462**, although other suitable securing arrangements, such as threads, may be utilized. The fixed spacer **666** includes interior threads **672** that receive exterior threads **678** of the adjustable spacer **668**. The fixed spacer **666** includes a plurality of detents **674** positioned on a helical portion of the fixed spacer **666**. The adjustable spacer **668** includes a spring detent arm **676** that engages one of the detents **674** to prevent rotation and axial displacement of the adjustable spacer **668** relative toward the fixed spacer **666**. The spring detent arm **676** is shaped and configured to pass over the detents **674** in one direction to allow rotation and axial displacement of the adjustable spacer **668** away from the fixed spacer **666**. The adjustable spacer **668** may be initially secured to the fixed spacer **666** via the threads **672**, **678** by applying a force to the top of the spring detent arm **676**, which biases the spring detent arm **676**

away from the detents **674** to allow the spacers **666**, **668** to be secured to each other. Accordingly, in the same manner as discussed above in connection with spacer assembly **460**, the adjustable spacer is free to rotate in one axial direction to adjust the length of the spacer assembly **660**. [0236] Referring again to FIGS. **94-100**, the spacer assembly **660** further includes a shim **680** configured to be received and secured to the adjustable spacer **668**. Rather than providing a plurality of sizes of adjustable spacers **468**, **668**, a plurality of shim **680** sizes can be provided to accommodate a plurality of different fill volumes within the container **14**. The shim **680** may be secured to the adjustable spacer **668** via a connector **682** extending from the shim **680** that is received by the adjustable spacer **668** using a snap-fit, although other suitable securing arrangements may be utilized. A center portion **684** of the fixed spacer **666** is configured to be engaged while the adjustable spacer **668** is rotated relative to the fixed spacer **666** to prevent rotation of the fixed spacer **666** along with the adjustable spacer **268**. The center portion **684** of the fixed spacer **666** is accessible through an opening in the shim **680**.

[0237] Elements of one disclosed aspect can be combined with elements of one or more other disclosed aspects to form different combinations, all of which are considered to be within the scope of the present invention.

[0238] While this disclosure has been described as having exemplary designs, the present disclosure can be further modified within the spirit and scope of this disclosure. This application is therefore intended to cover any variations, uses, or adaptations of the disclosure using its general principles. Further, this application is intended to cover such departures from the present disclosure as come within known or customary practice in the art to which this disclosure pertains and which fall within the limits of the appended claims.

Claims

1. A drug delivery system comprising: a container configured to hold a medicament; a valve assembly; a stopper configured to move within the container to dispense the medicament from the container; and a drive assembly configured to move the stopper within the container, the drive assembly comprising: a first plunger member configured to move axially between a pre-use position, a use position, and a post-use position; a second plunger member received by the first plunger member; a first biasing member received by the second plunger member and configured to move the stopper within the container; and a second biasing member received by the second plunger member and configured to drive the first plunger member against the stopper to move the container into engagement with the valve assembly, wherein the second plunger member is configured to move axially after the first plunger member has moved a predetermined axial distance.
2. The drug delivery system according to claim 1, wherein the drive assembly further comprises a plunger actuation member configured for engagement and release from the first plunger member.
3. The drug delivery system according to claim 2, wherein the plunger actuation member is rotationally moveable relative to the first plunger member between a first rotational position and a second rotational position.
4. The drug delivery system according to claim 3, wherein the first rotational position is 15 degrees from the second rotational position.
5. The drug delivery system according to claim 2, wherein the plunger actuation member comprises a plurality of projections configured to be received by a plurality of slots or cutouts defined by the first plunger member.
6. The drug delivery system according to claim 5, wherein, in a first rotational position of the plunger actuation member, the plurality of projections and the plurality of slots or cutouts are out of alignment such that the plunger actuation member is engaged with the first plunger member to prevent movement of the first plunger member and the second plunger member, with the first

biasing member and the second biasing members biasing the first plunger member and the second plunger member away from the plunger actuation member.

7. The drug delivery system according to claim 6, wherein, in a second rotational position of the plunger actuation member, the plurality of projections and the plurality of slots or cutouts are aligned with each other such that the plunger actuation member is disengaged with the first plunger member to allow movement of the first plunger member and the second plunger member.

8. The drug delivery system according to claim 2, further comprising a needle actuator assembly configured to engage a drive surface of the plunger actuation member to move the plunger actuation member from the first rotational position to the second rotational position.

9. The drug delivery system according to claim 8, wherein the drive surface comprises a plurality of gears.

10. The drug delivery system according to claim 1, wherein the first plunger member forms a telescoping arrangement with the second plunger member.

11. The drug delivery system according to claim 1, wherein each of the first biasing member and the second biasing member comprises a compression spring.

12. The drug delivery system according to claim 1, wherein the first biasing member is received by the second plunger member and is constrained between the plunger actuation member and a first spring seat of the second plunger member.

13. The drug delivery system according to claim 1, wherein the second biasing member is constrained between a second spring seat of the second plunger member and the first plunger member.

14. The drug delivery system according to claim 1, wherein the second biasing member is stiffer than the first biasing member.

15. A drug delivery system comprising: a container configured to hold a medicament; a needle; a valve assembly configured to pierce a closure of the container to place the container in fluid communication with the needle; a stopper configured to move within the container to dispense the medicament from the container to the needle; and a drive assembly configured to move the stopper within the container, the drive assembly comprising: a first plunger member configured to move axially between a pre-use position, a use position, and a post-use position; a second plunger member received by the first plunger member; a plunger actuation member configured for engagement and release from the first plunger member; a first biasing member positioned between the second plunger member and the plunger actuation member, the first biasing member configured to move the stopper within the container; and a second biasing member positioned between the second plunger member and the first plunger member, the second biasing member configured to drive the first plunger member against the stopper to move the container into engagement with the valve assembly, wherein the second plunger member is configured to move axially after the first plunger member has moved a predetermined axial distance.

16. The drug delivery system according to claim 15, wherein the plunger actuation member is rotationally moveable relative to the first plunger member between a first rotational position and a second rotational position.

17. The drug delivery system according to claim 16, wherein the plunger actuation member comprises a plurality of projections configured to be received by a plurality of slots or cutouts defined by the first plunger member.

18. The drug delivery system according to claim 17, wherein, in the first rotational position of the plunger actuation member, the plurality of projections and the plurality of slots or cutouts are out of alignment such that the plunger actuation member is engaged with the first plunger member to prevent movement of the first plunger member and the second plunger member, with the first biasing member and the second biasing members biasing the first plunger member and the second plunger member away from the plunger actuation member.

19. The drug delivery system according to claim 18, wherein, in the second rotational position of

the plunger actuation member, the plurality of projections and the plurality of slots or cutouts are aligned with each other such that the plunger actuation member is disengaged with the first plunger member to allow movement of the first plunger member and the second plunger member.
